Cervical premalignancy and colposcopy

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Pre-invasive disease of the lower genital tract

- Preinvasive disease of the female lower genital tract encompasses a spectrum of intraepithelial neoplasia of the cervix, vulva and vagina. These changes are referred to as cervical, vulval and vaginal intraepithelial neoplasia (CIN, VIN and VAIN, respectively).

Epidemiology

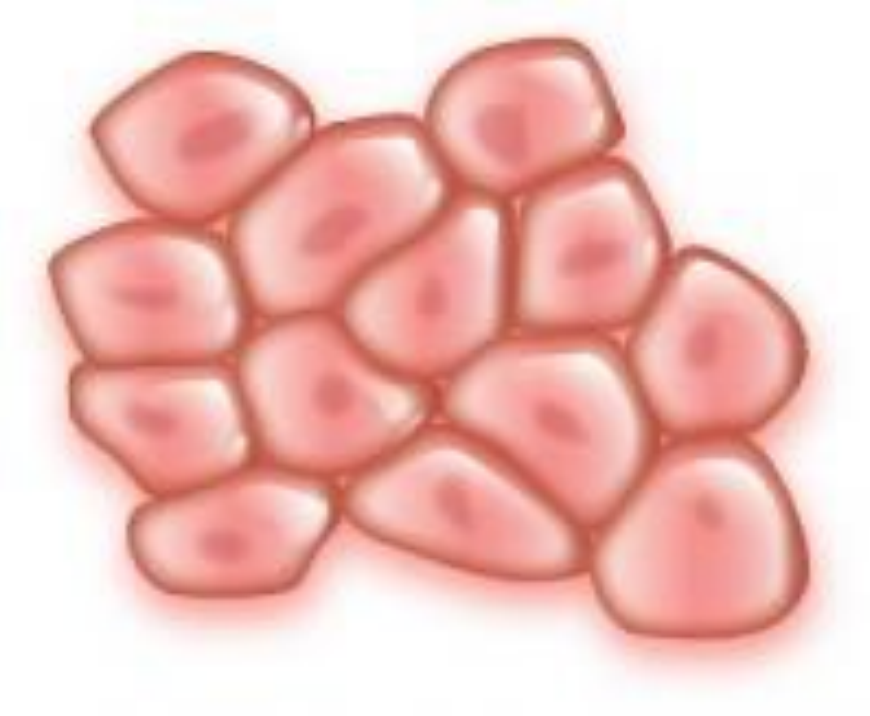
- At least 530 000 new cases of cervical cancer are diagnosed worldwide each year; and 80% of these cases occur in the developing world.
- Worldwide cervical cancer is the third most common cancer among women and causes more deaths than any other cancer – approximately one death every 2 minutes.
- .The estimated annual incidence of CIN among women who undergo cervical cancer screening is 4% for CIN1, and 5% for CIN2 and CIN3.
- Cancer of the vulva is rare and, when coupled with cancer of the vagina, accounts for less than 1% of all cancer cases and 6% of gynaecological cancers diagnosed in women in the UK.

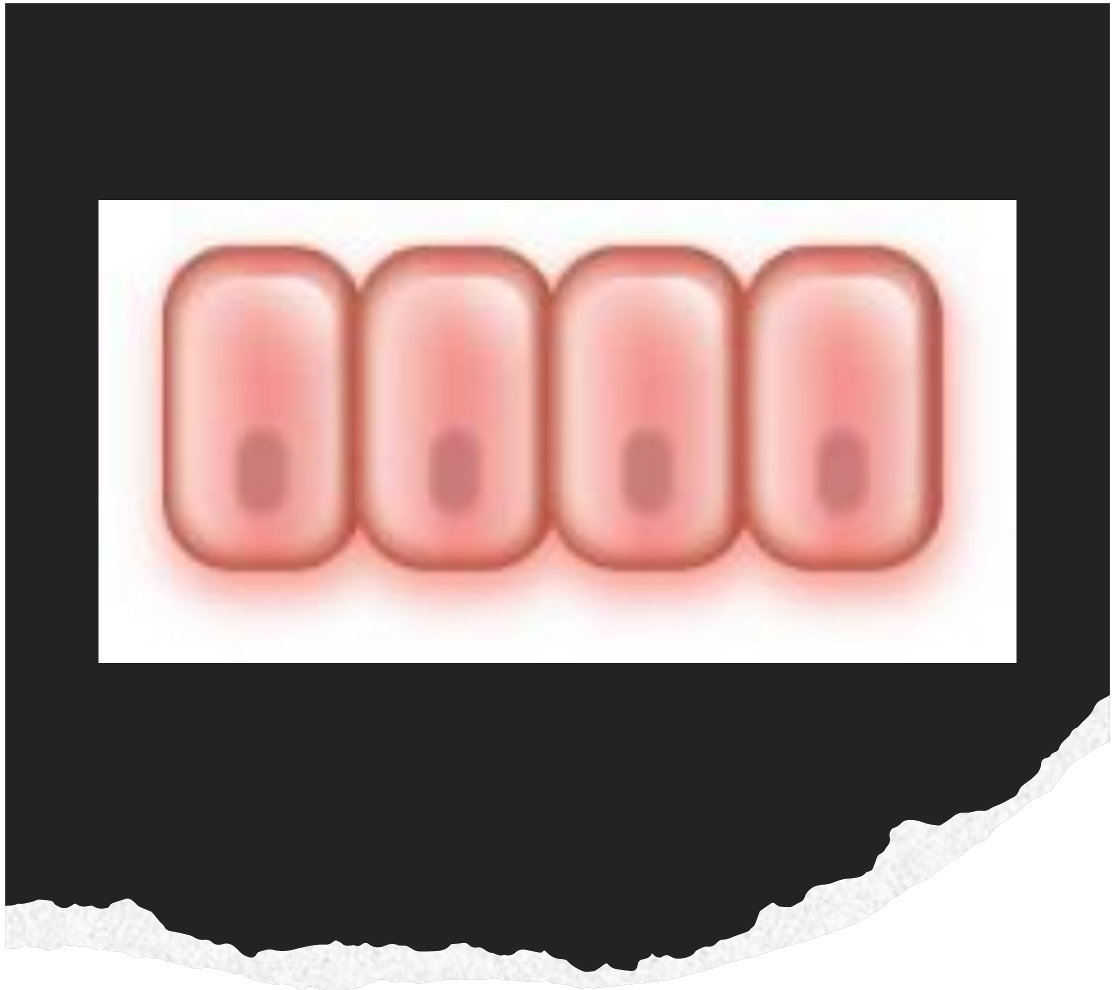
Risk factors

- The following factors are associated with an increased risk of genital HPV infection:
- Number of sexual contacts of male and/or female partner (higher number of partners = higher risk)
- persistent infection
- Higher viral load
- Immune compromise
- Cigarette smoking (women who smoke are twice as likely to develop cervical cancer than non-smokers)
- low socioeconomic status
- prolonged use of oral contraceptive pill (OCP; note that overall, the benefits of taking the OCP far outweigh the increased risk of HPV infection and malignancy)
- Number of pregnancies (higher number of pregnancies = higher risk).
- It should be noted that:
- Using a condom gives some protection against HPV infection
- A late first pregnancy lowers the risk.

Histology of the cervix

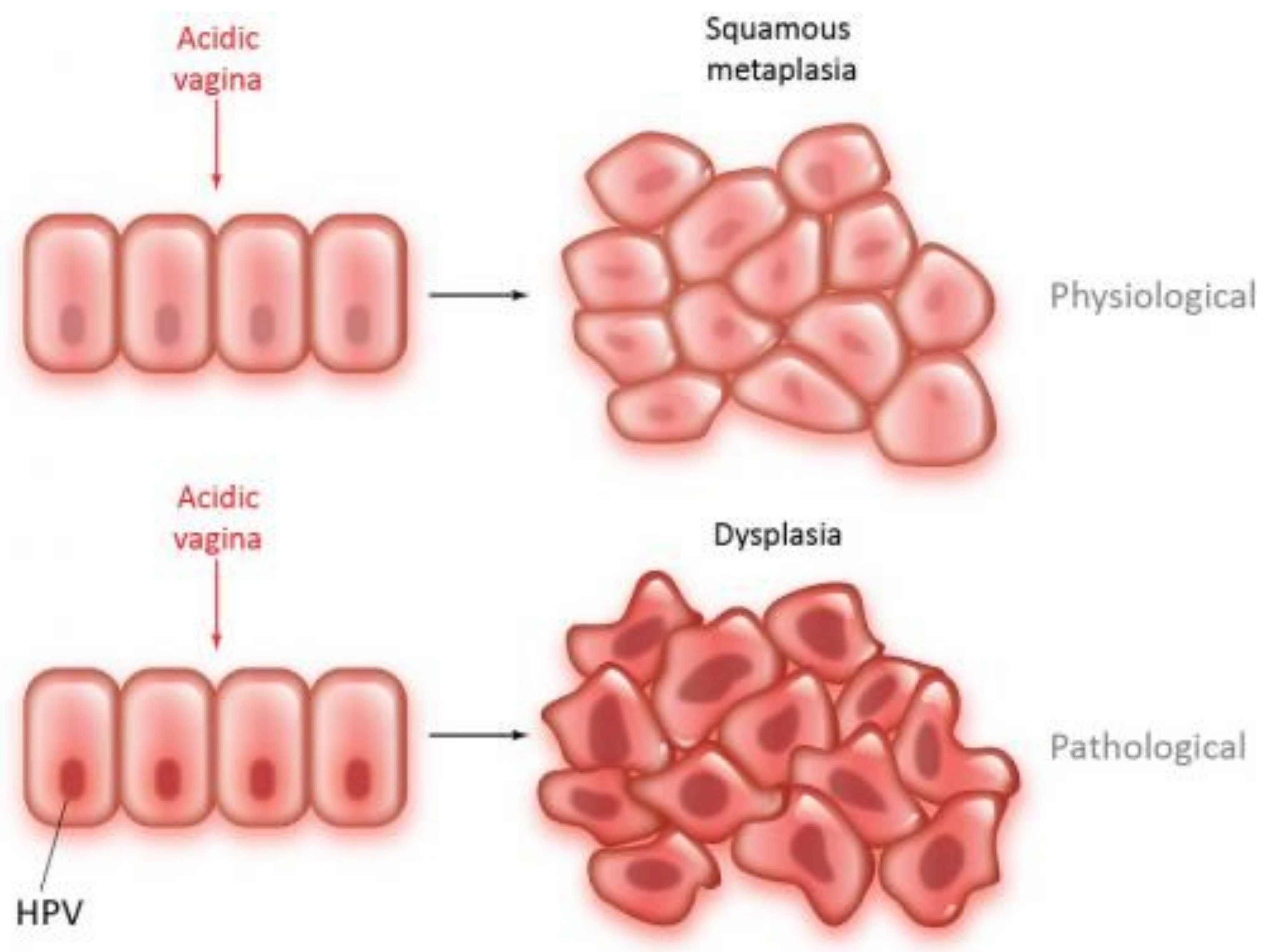
- **The ectocervix**
- The ectocervix is non-keratinising stratified *squamous* epithelium.
- It is resistant to low vaginal pH.

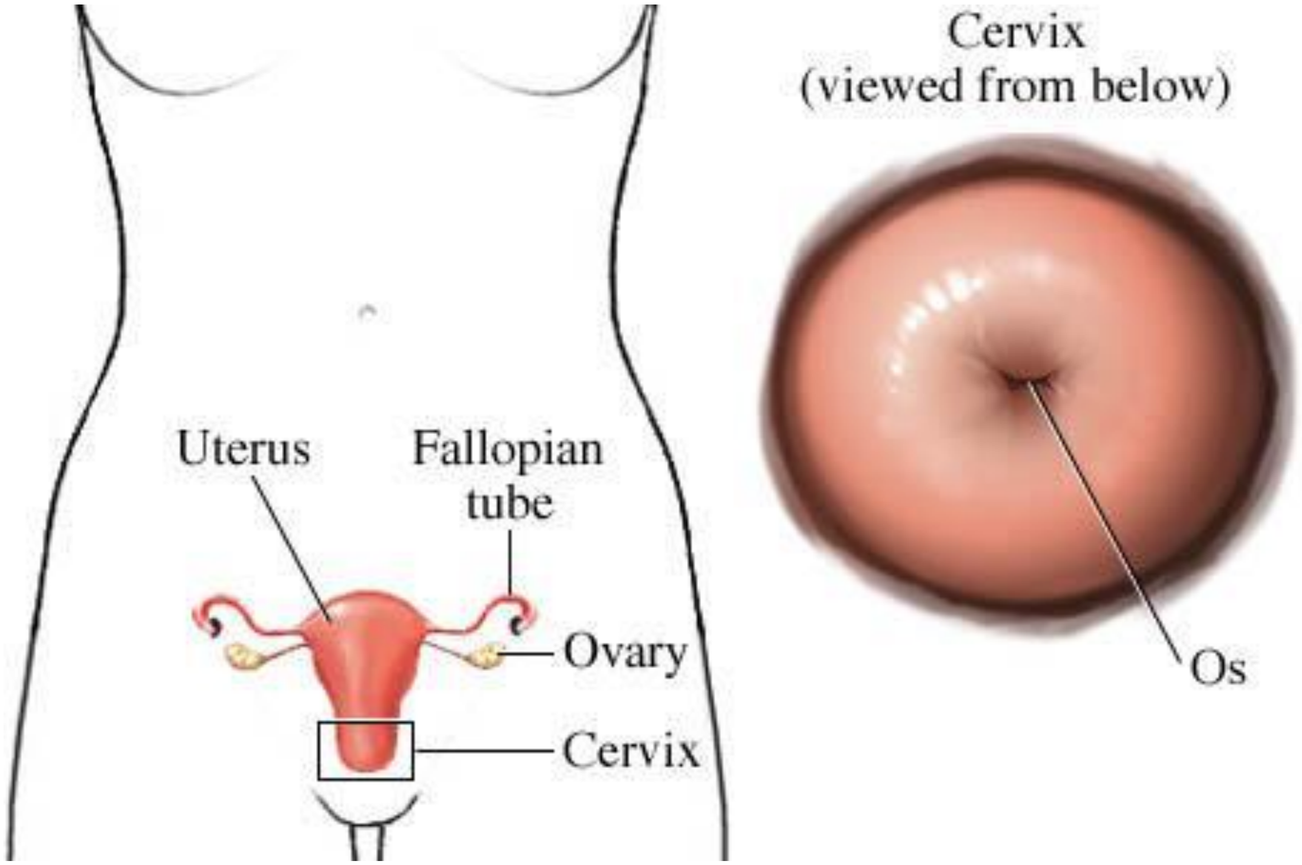


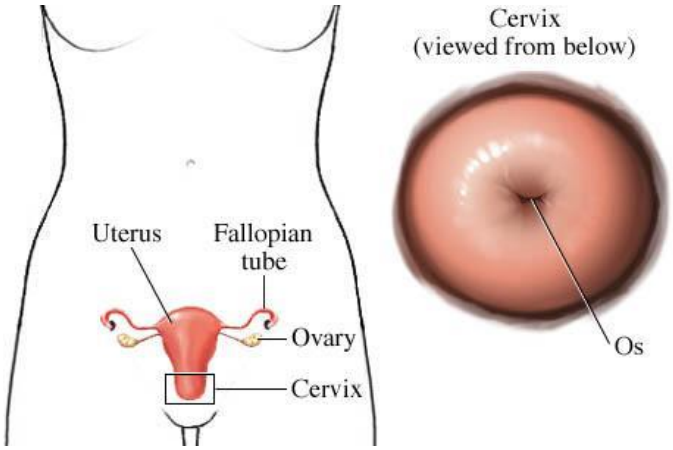


Dysplastic change

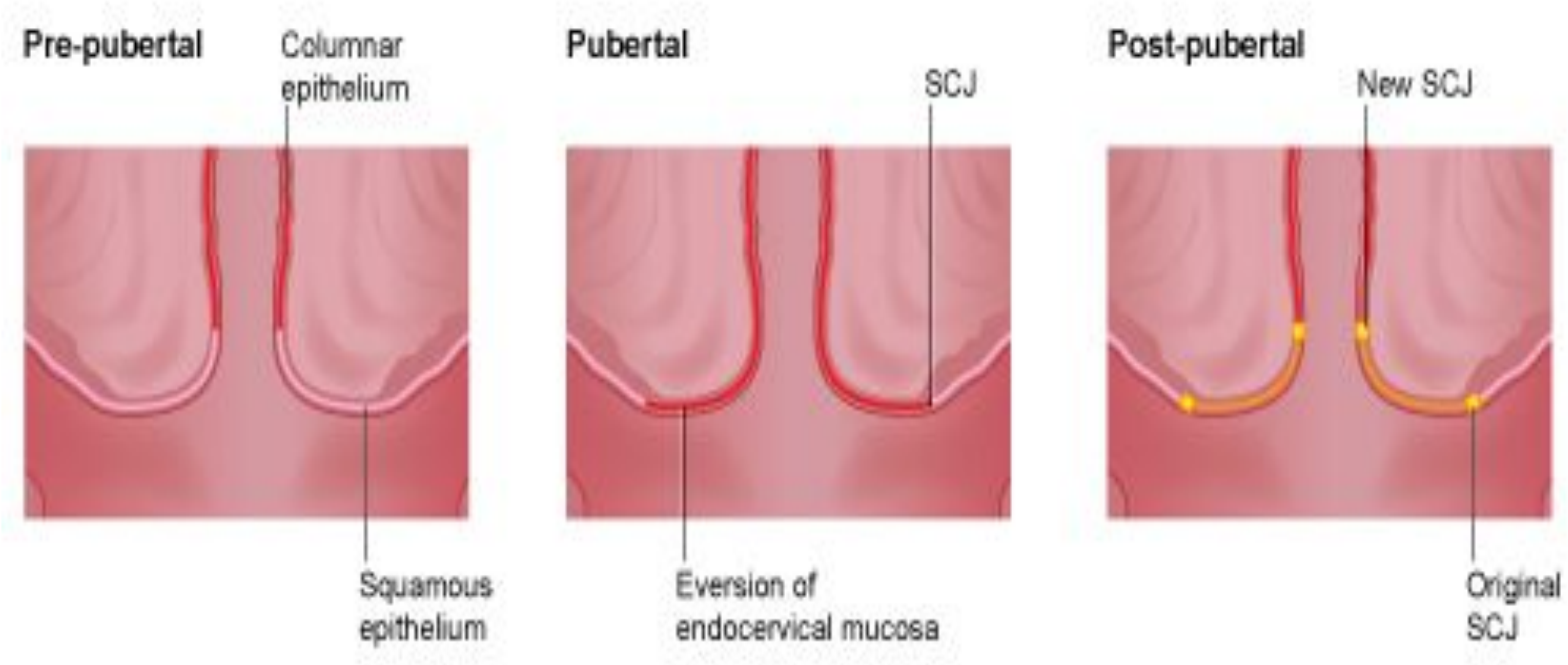
If cells are infected with HPV then physiological *metaplastic* change can become disturbed such that the cells undergo dysplastic change:




Cervix
(viewed from below)
Uterus
Fallopian tube
Ovary
Cervix
Os

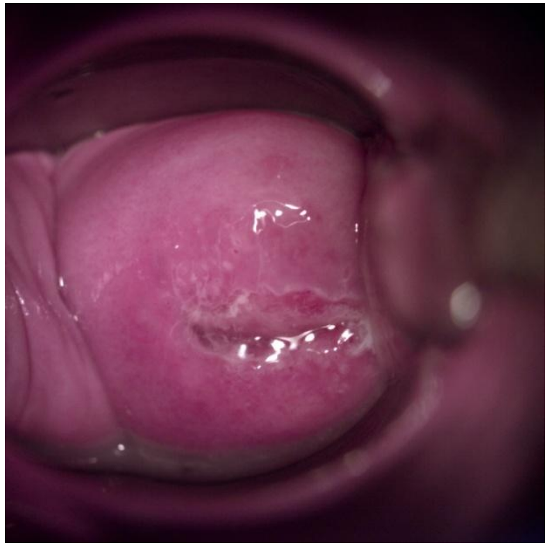

Cervix
(viewed from below)
Uterus
Fallopian tube
Ovary
Cervix
Os

The junction between ectocervix and endocervix is the squamocolumnar junction. From late fetal life to the menopause the position of this junction changes:



Squamocolumnar Junction (SCJ)

- A visible line of demarcation
- Between endocervical columnar and squamous or metaplastic squamous epithelium
- Either on the ectocervix or with in the cervical canal



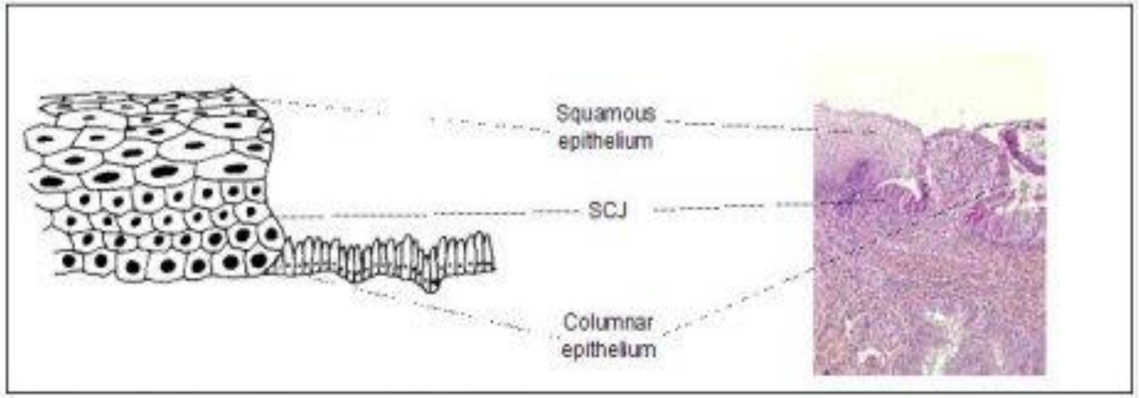
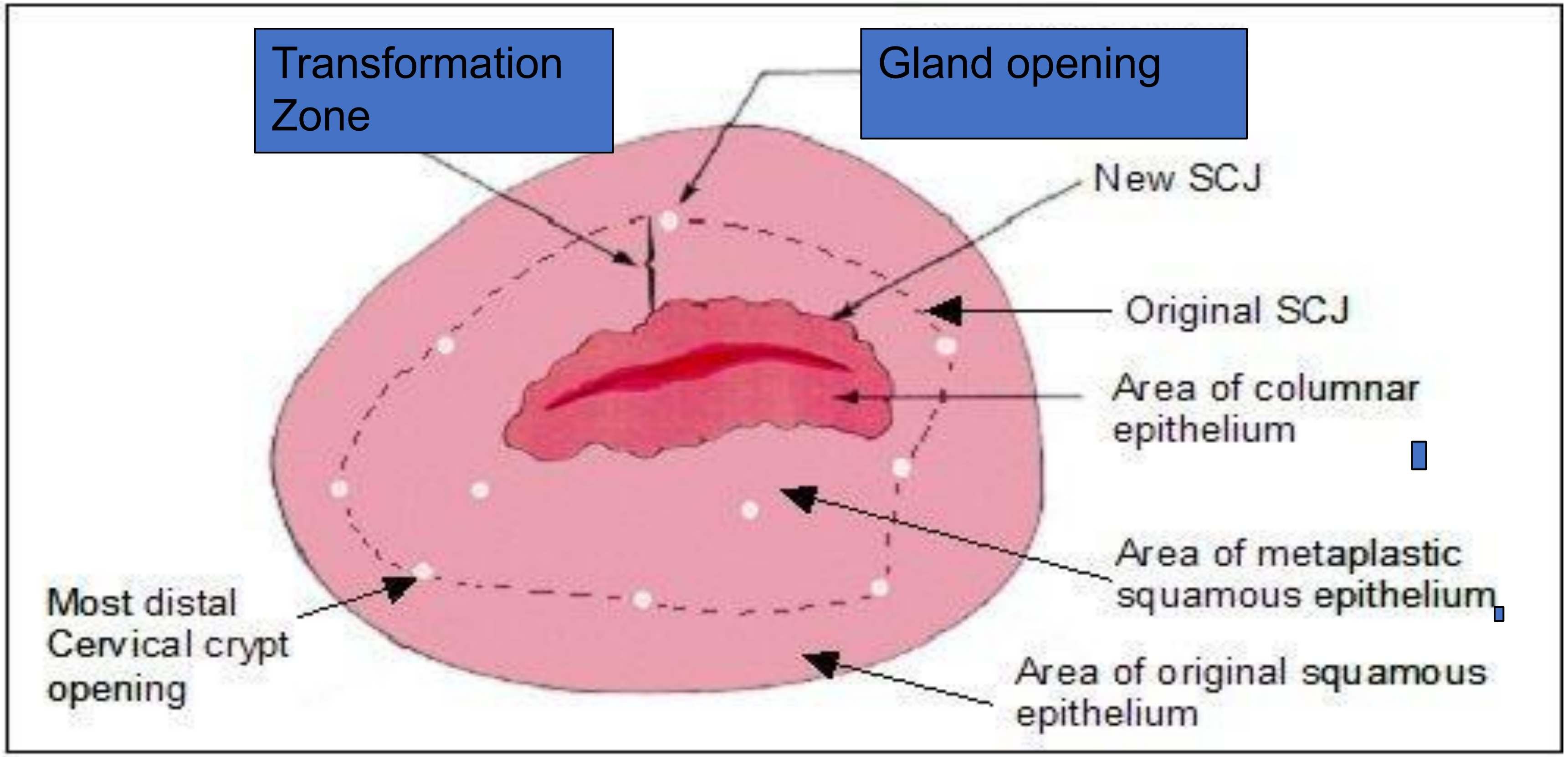


FIGURE 1.6: Squamocolumnar junction (SCJ) (×10)

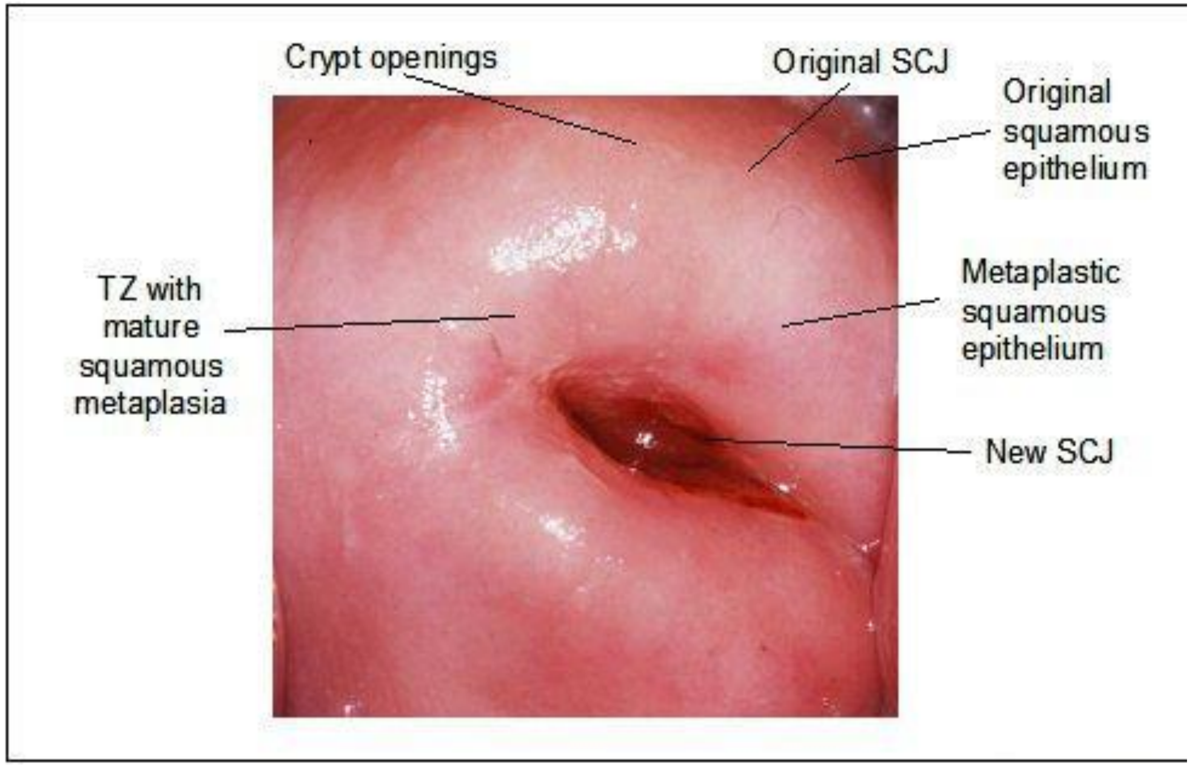
Transformation Zone

- The topographic area between the original SCJ and the current SCJ
- May contain gland openings and islands of columnar epithelium surrounded by squamous metaplasia

Transformation Zone



Transformation Zone



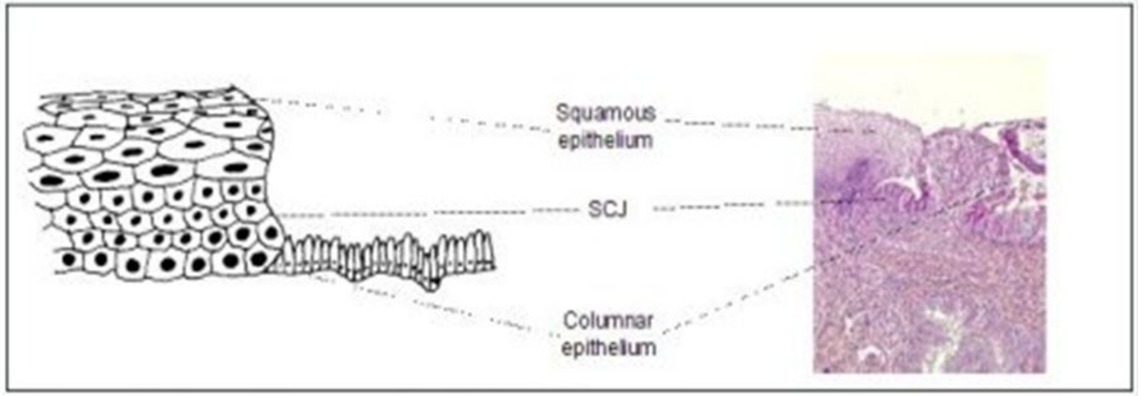


FIGURE 1.6: Squamocolumnar junction (SCJ) (×10)

Transformation zone classification

Type	Description
Type 1	Ectocervical Entire transformation zone visible
Type 2	Endocervical component Entire transformation zone visible (use of endocervical forceps may facilitate this)
Type 3	Endocervical component Entire transformation zone not visible

Pathology of the cervix – cytology

Dyskaryosis

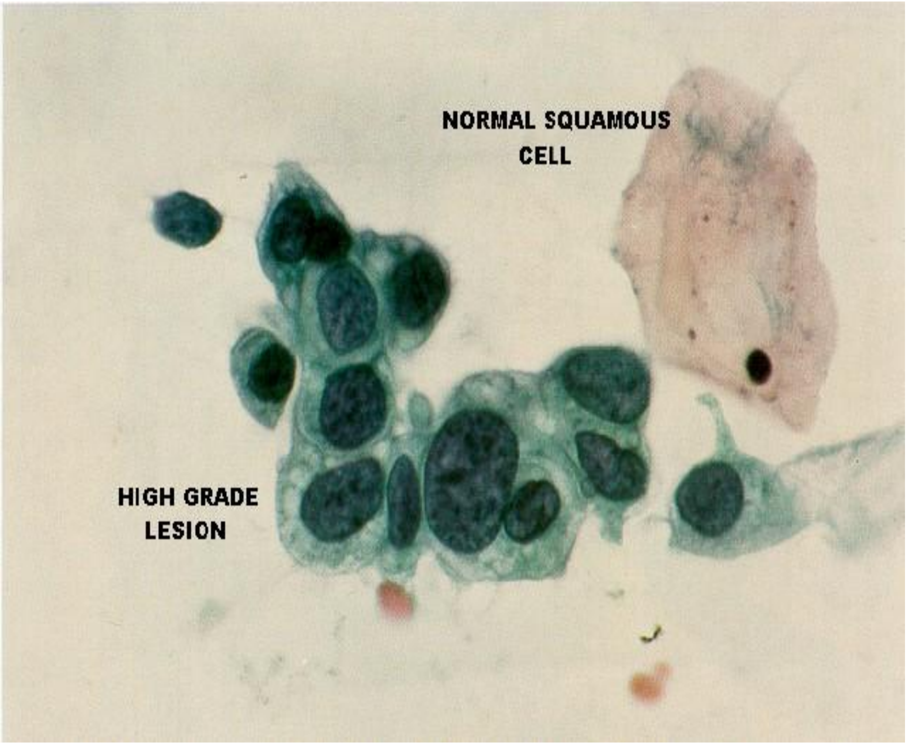
- Dyskaryosis is a *cytological* diagnosis:
- Increased nuclear: cytoplasmic ratio
- Mitotic figures
- Nuclear pleomorphism
- Nuclear hyperchromasia

Human papillomavirus

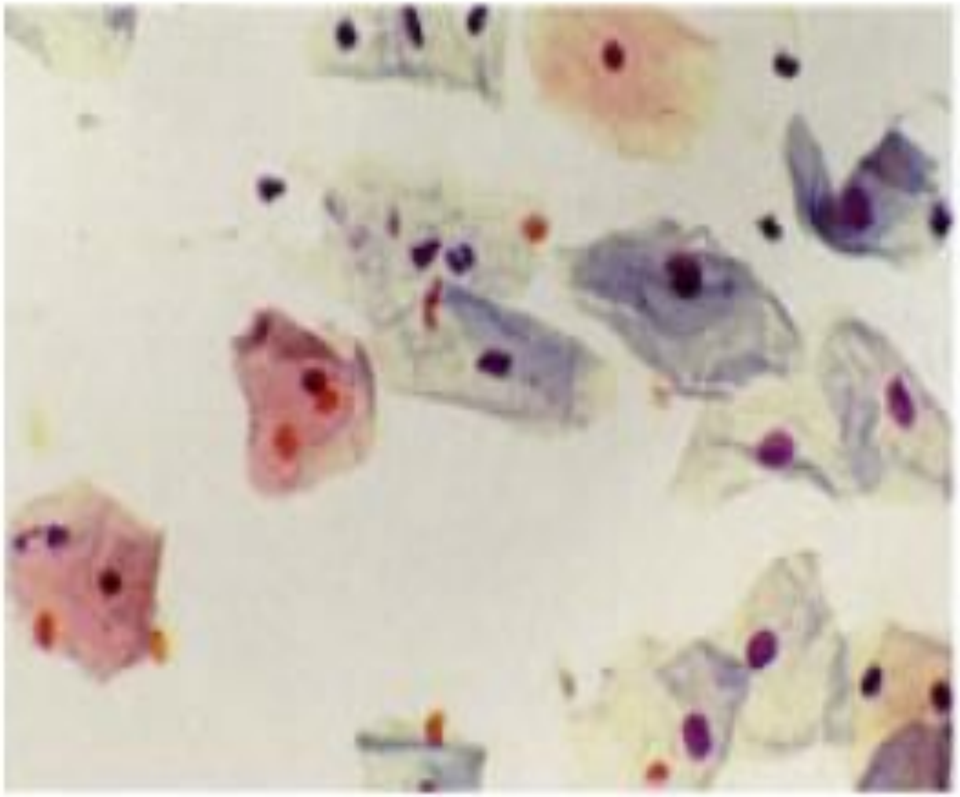
- Features associated with HPV include:
- Koilocytosis (vacuolated cells)
- Hyperkeratosis
- Multinucleation.

Pre-Malignant Changes

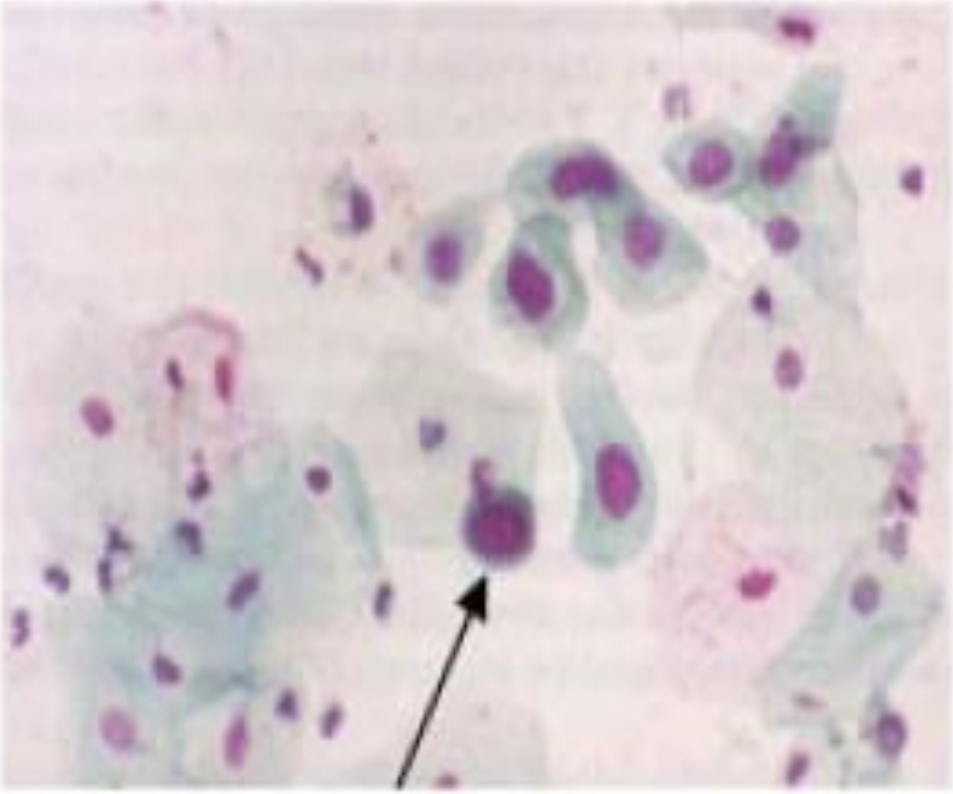
- CIN = cervical intraepithelial neoplasia
- Large abnormal nuclei
- Reduced cell cytoplasm
- loss of cell stratification and maturation throughout thickness of the endothelium



Normal smear

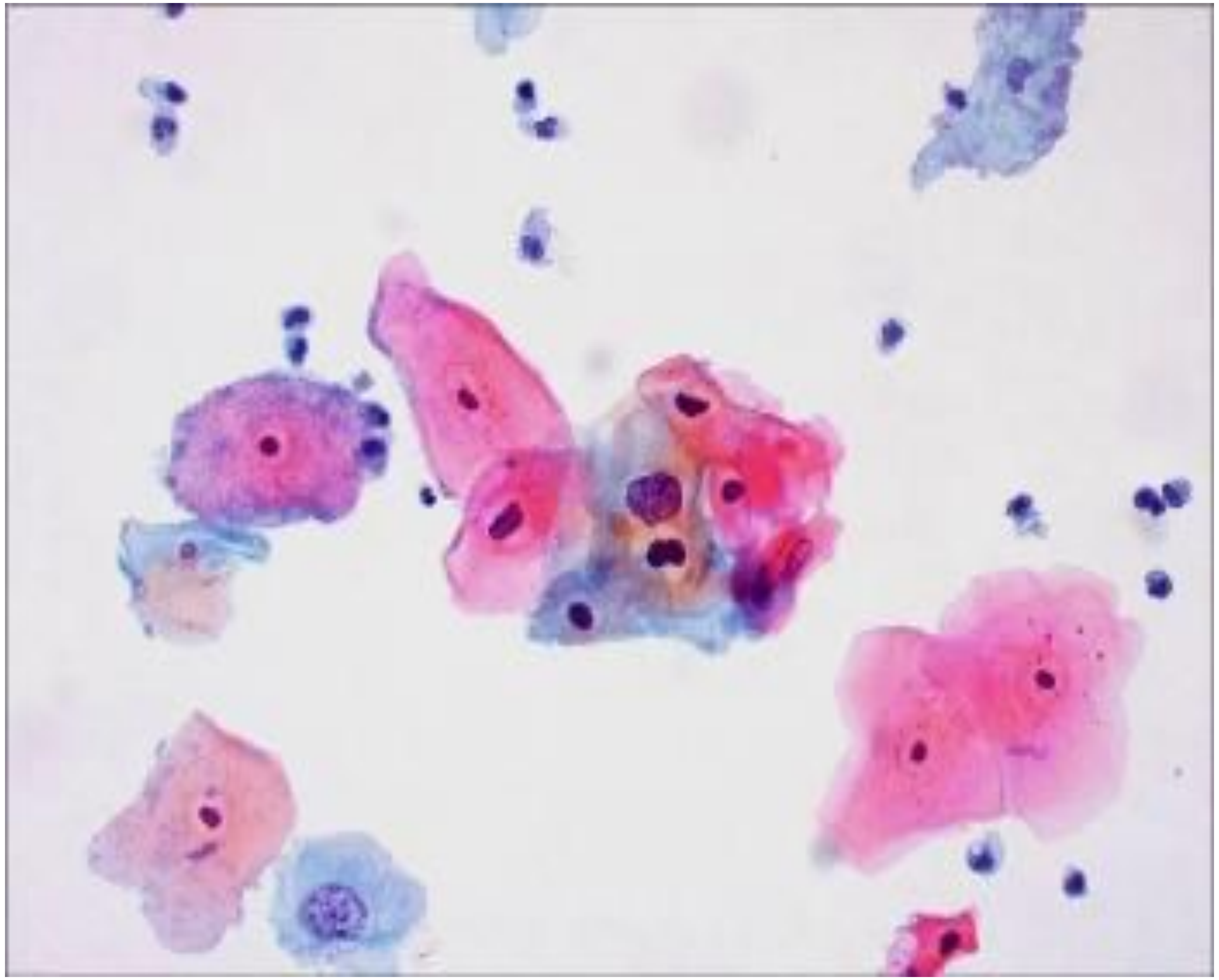


Severe dyskaryosis



Increased nuclear:cytoplasmic ratio with nuclear hyperchromasia

Cytopathological specimen of mild dyskaryosis.



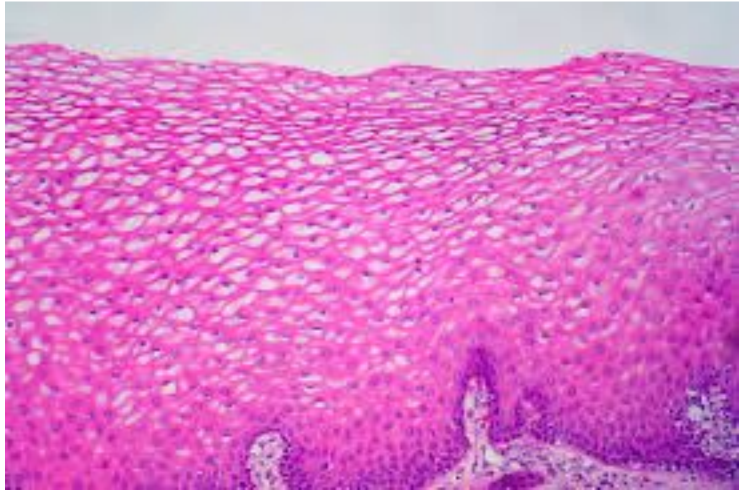
Pathology of the cervix – histology

Stage	Description
CIN1	Changes most apparent in basal third of epithelium ± HPV change
CIN2	Changes affecting basal two-thirds of epithelium, more marked nuclear atypia
CIN3	More severe changes affecting the entire epithelium

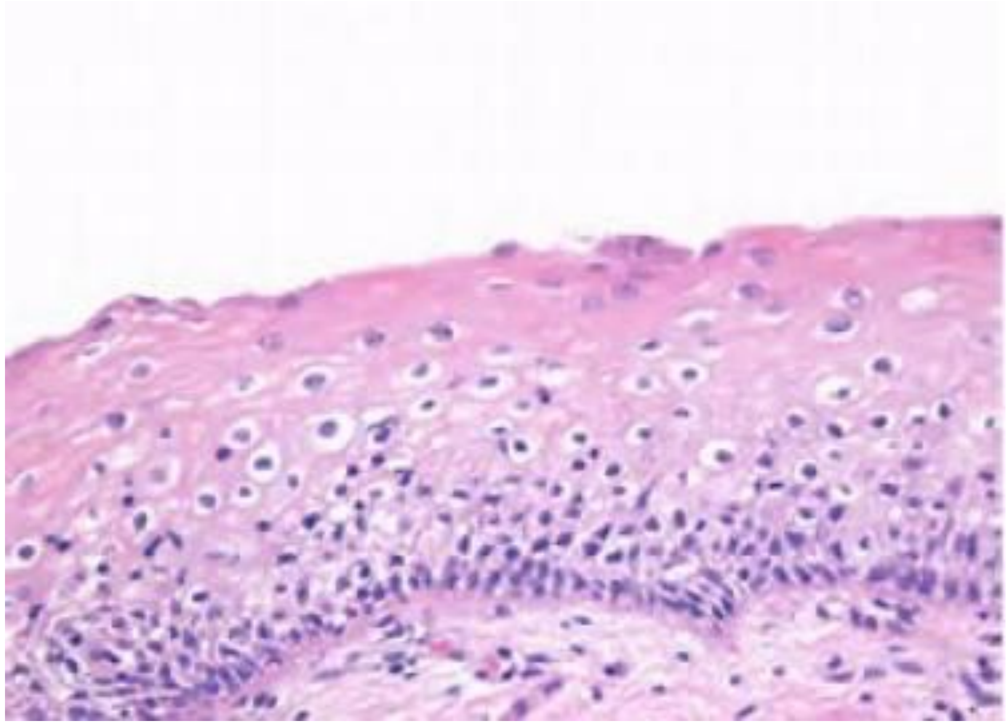
Terminology Of CIN

CIN 1		Mild Dysplasia
CIN 2		Moderate dysplasia
CIN 3		Severe dysplasia
		Carcinoma *in situ*

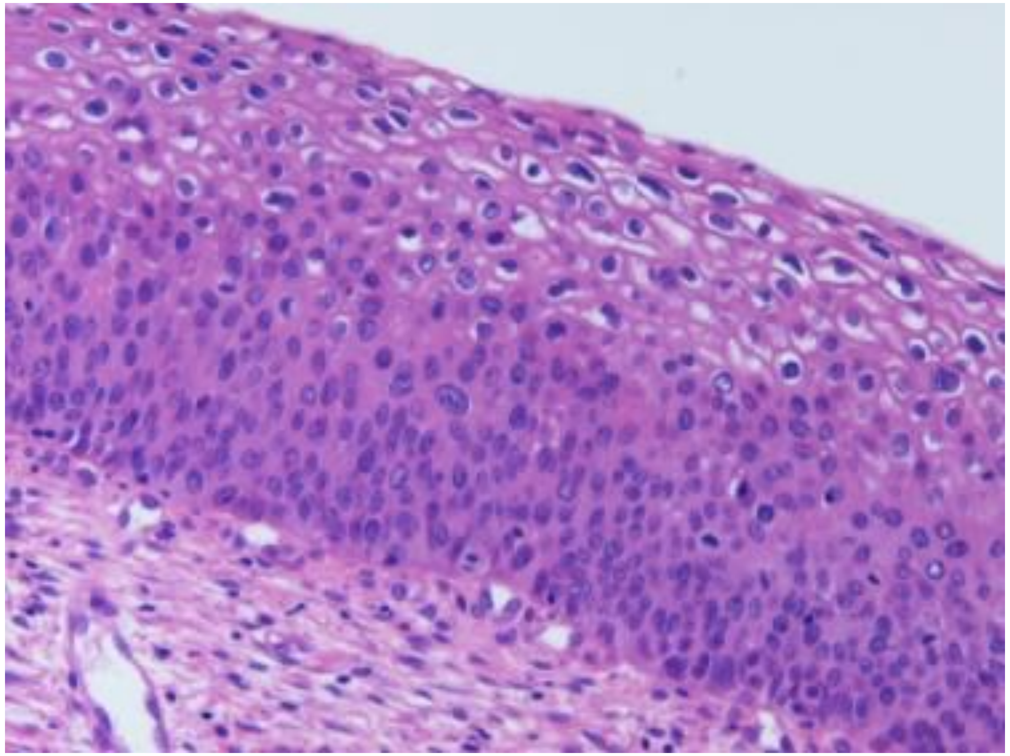
Histopathological specimen of CIN1. Copyright Ed Uthman, MD



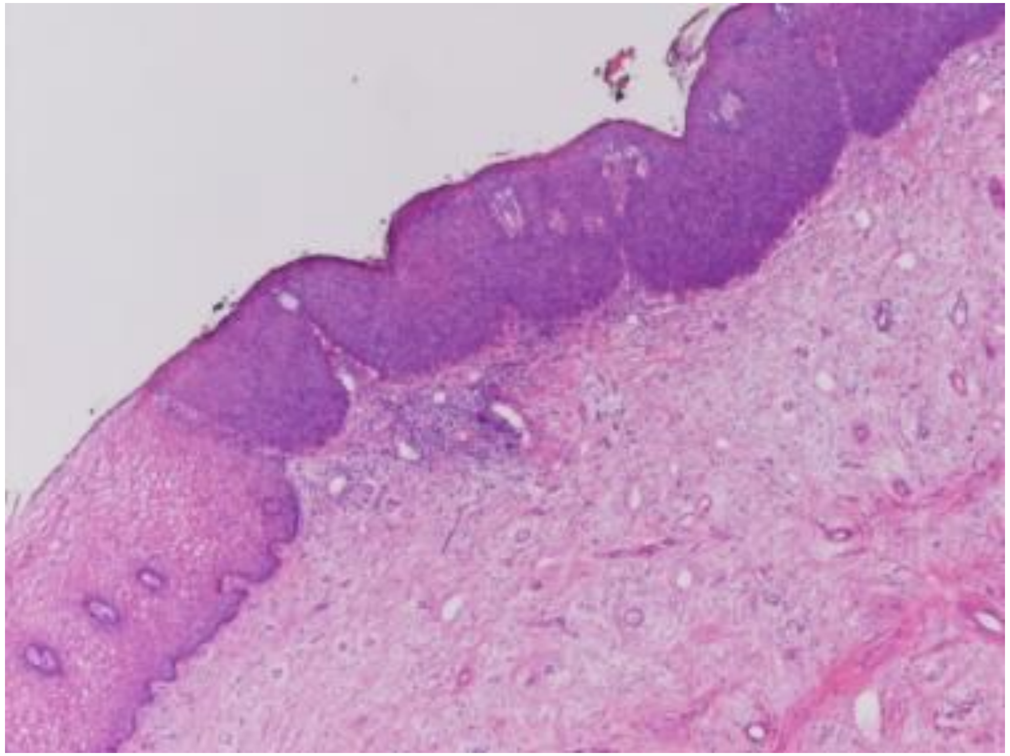
CIN 1



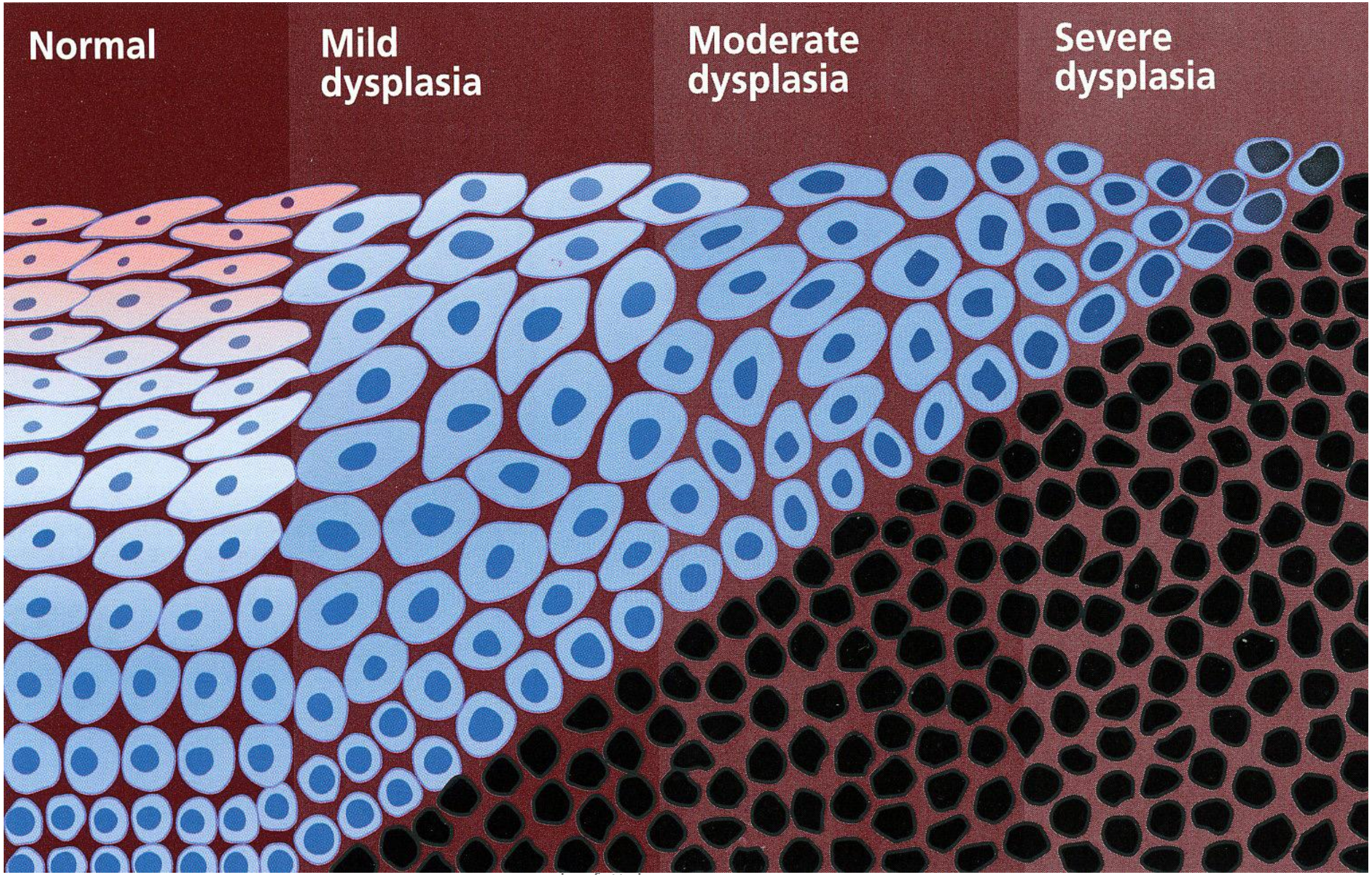
CIN2



CIN 3



CIN



Normal
Mild dysplasia
Moderate dysplasia
Severe dysplasia

Natural history

	CIN1	CIN2	CIN3
Regress	57%	43%	32%
Persist	32%	35%	56%
Progress	11%	22%	12%
Invasive carcinoma	1%	5%	>12%

Data from Ostör AG (1993)(link is external).

Not all CIN will progress to CIN3. The majority of low-grade disease spontaneously resolves within 24 months as the patient develops immunity to HPV and expels the virus. Some authors believe that most sexually active women have CIN at some stage, but that it is transient.

- The use of the cervical smear test as a screening tool for detecting pre-malignant disease of the cervix has been highly successful. In the UK, there has been a greater than a 20% reduction in the incidence of squamous cervical cancer since 1986, with mortality rates falling by 7% a year.
- It is estimated that cervical screening currently saves 2000 lives a year, and it is regarded generally as a cost-effective public health measure. It is only effective if the test is offered in an organised way to all women at risk, Quality control measures are also important to maintain standards.
- Despite the success of organised population cervical screening, a negative smear does not confer 100% protection from cervical cancer, as the sensitivity of the test is approximately 70%. This results in at least 10% false-negative smears. A combination of factors contributes to these errors, the majority arising from sampling or slide preparation errors, with inaccurate reporting accounting for the rest.

In the UK, the NHS Cervical Screening Programme screens:

- women between 25 and 49 years every 3 years
- women between 49 and 64 years every 5 years.
- Women less than 25 years of age are generally not screened, as changes in the cervix in this age group will usually resolve and detection of these changes may lead to overtreatment with its consequences. Furthermore, screening at a younger age will detect a high degree of low-grade smears, which would significantly increase workload.
- When the woman is determined to be high-risk HPV positive, she is referred for a colposcopy, and women who are high-risk HPV negative are returned to routine recall. When high-grade dyskaryosis or worse is detected, referral for a colposcopy without a HR-HPV is done. In general, HR-HPV testing detects more than 90% of all cases of CIN2, CIN3 and invasive cancer, and is 25% more sensitive than liquid-based cytology in detecting borderline changes or worse.

Colposcopy Mashine

Colposcopic examination

- Colposcopic examination involves magnified stereoscopic visualisation of the cervix. It allows a clinical opinion to be formed and facilitates directed tissue sampling.
- For colposcopic examination of the cervix to be satisfactory, the entire transformation zone and the full extent of abnormal (atypical) epithelium is visible must be visible.
- If the new squamocolumnar junction can be seen colposcopically, then the entire transformation zone can be visualised.

Colposcopy

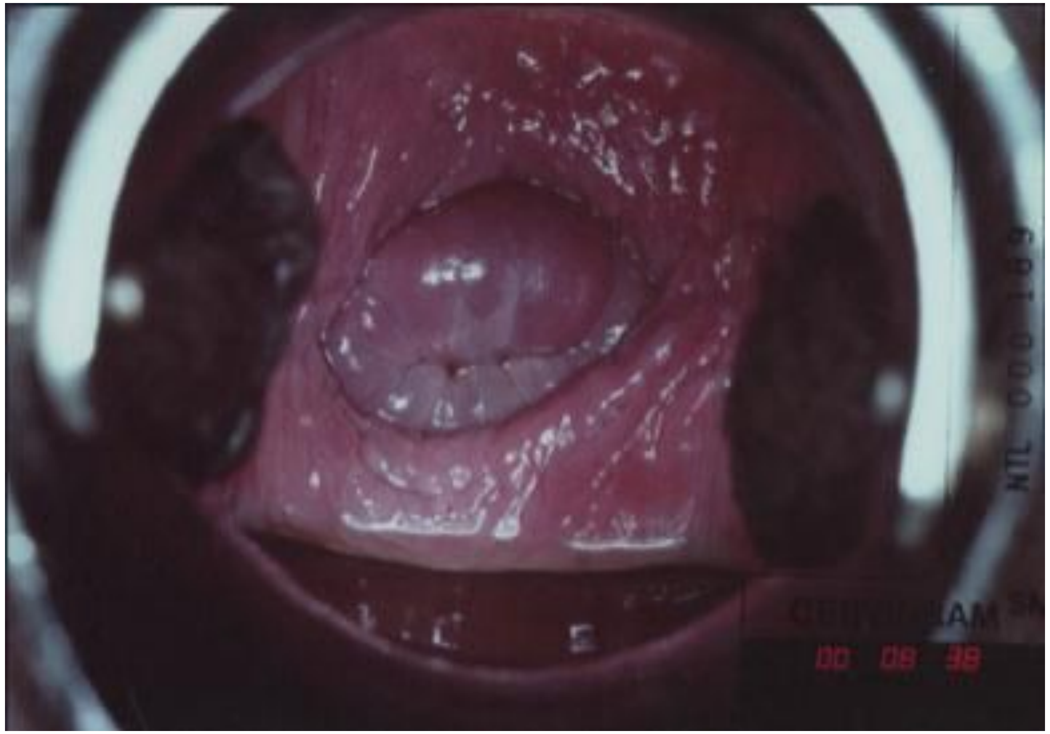
Colposcopy referral guidelines

Women should be referred for colposcopy if they have:

- Three consecutive inadequate smears
- Borderline nuclear abnormality (squamous or glandular) with high-risk HPV positive
- One mild dyskaryosis with high-risk HPV positive
- One moderate dyskaryosis
- One severe dyskaryosis
- One smear with possibility of invasion
- One smear with possibility of glandular neoplasia.
- After treatment (by loop or thermocoagulation) if high-risk HPV positive (irrespective of cervical smear result).

Colposcopy continued

- The severity of the lesion is correlated with:
- the intensity of acetowhite change
- sharp margins
- atypical blood vessels.



CIN1 after application of acetowhite

Mosaicism / Punctation

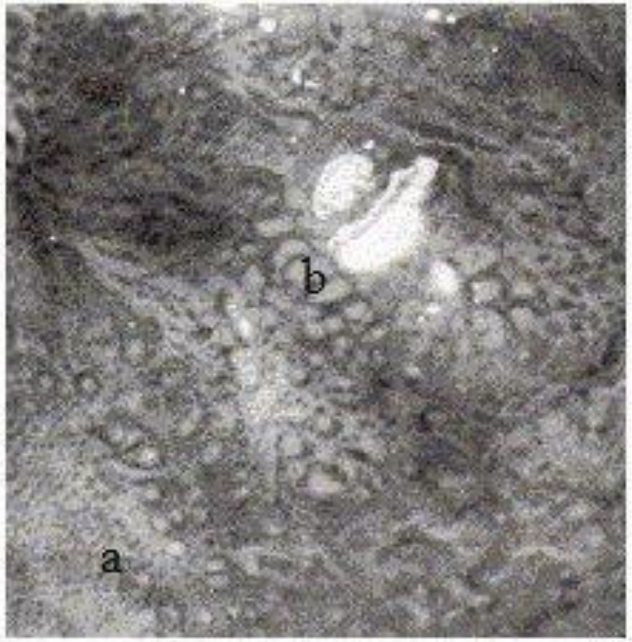


FIGURE 7.2a: Fine punctation (a) and coarse mosaic (b) seen afterapplication of normal saline.

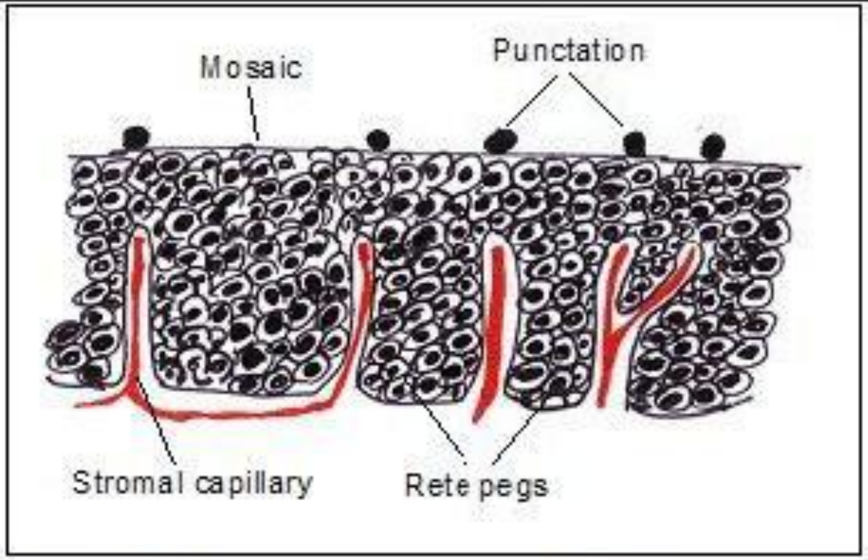
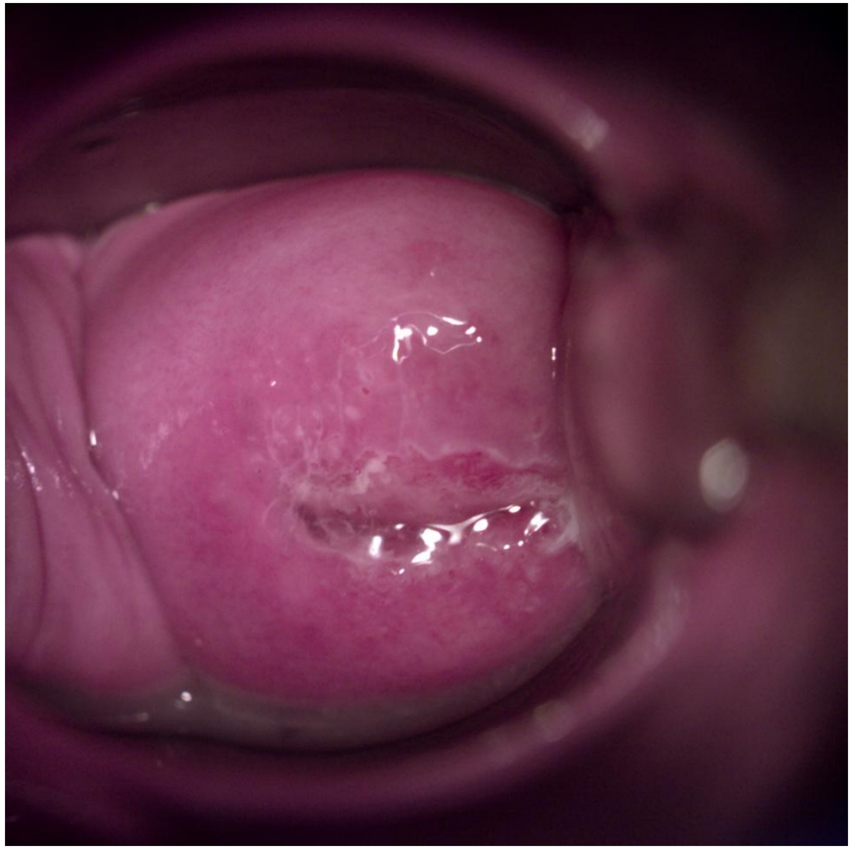
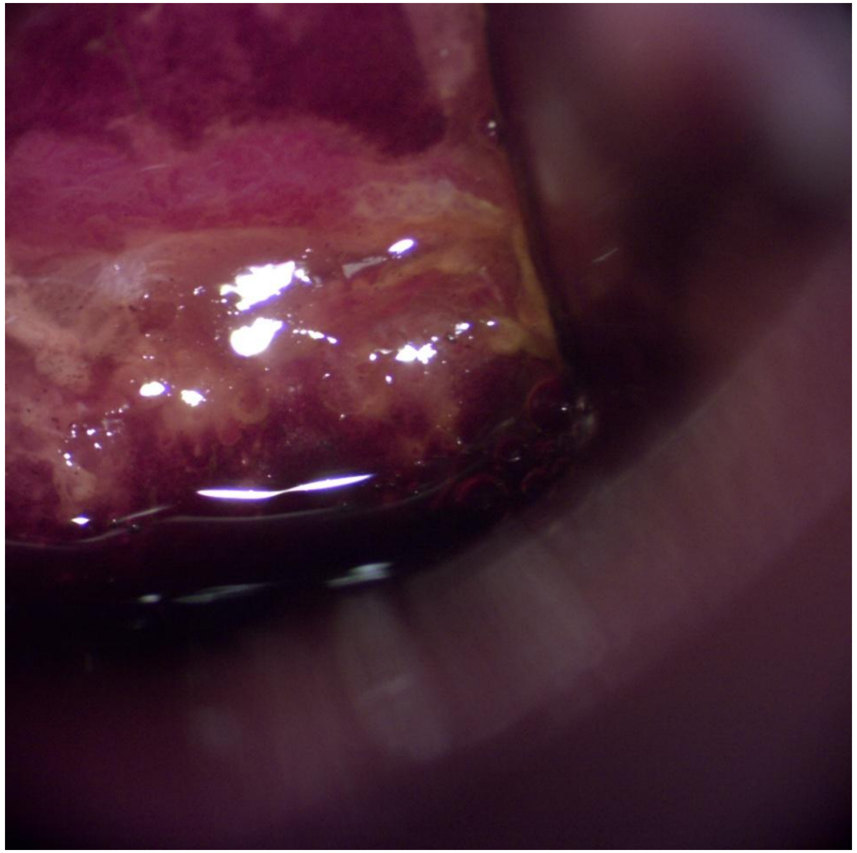


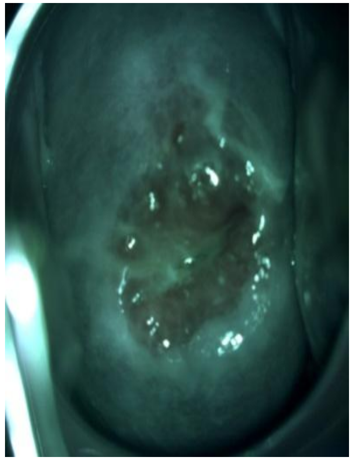
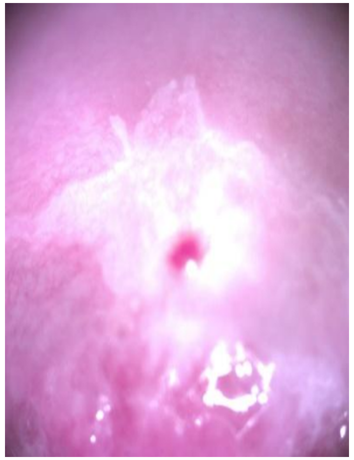
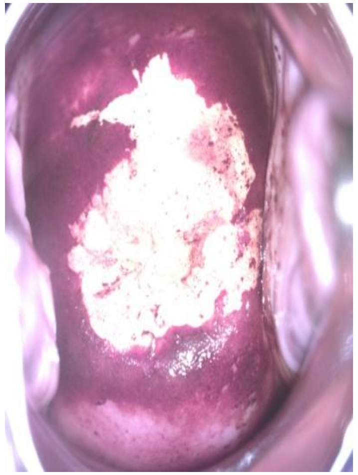
FIGURE 7.2b: Schematic diagram to show the rete pegs and the stromal capillaries which on end on view appear as punctations.

Original Squamous Epithelium

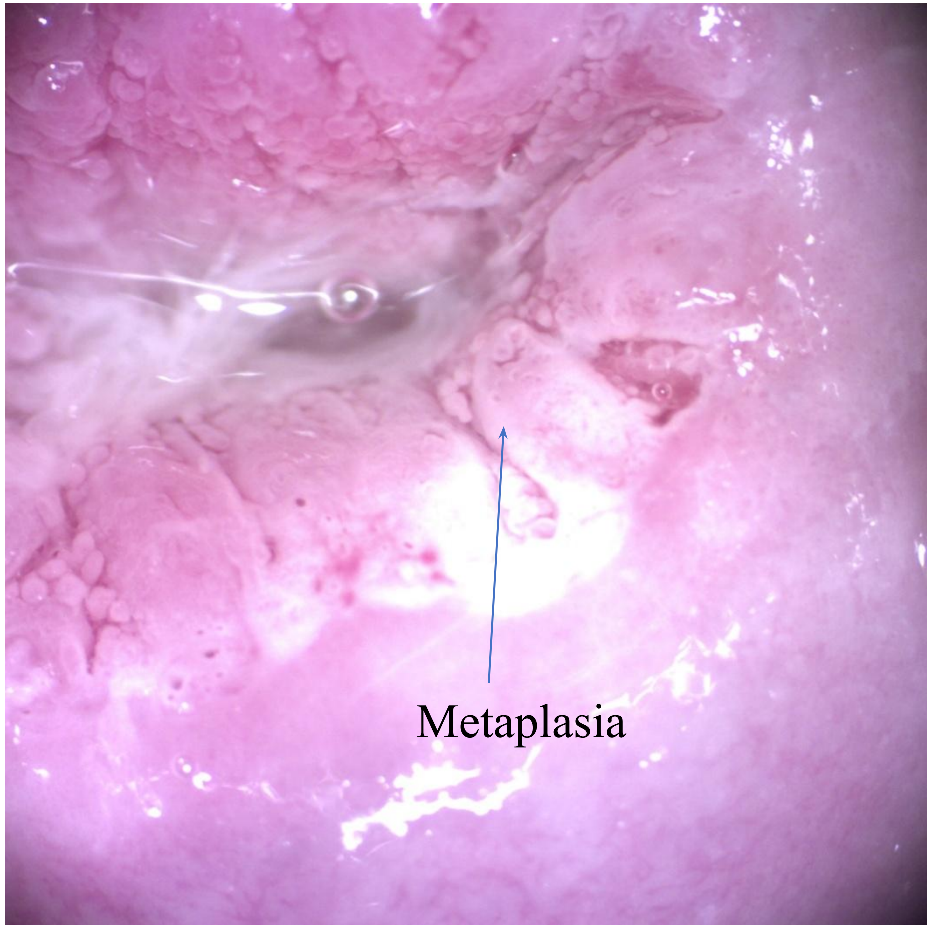


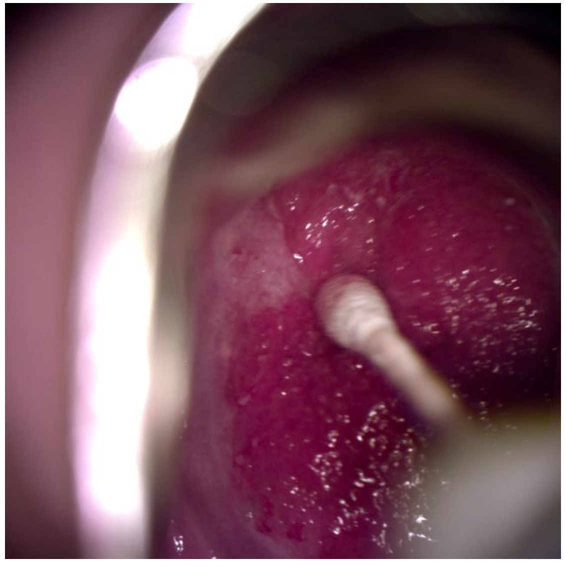


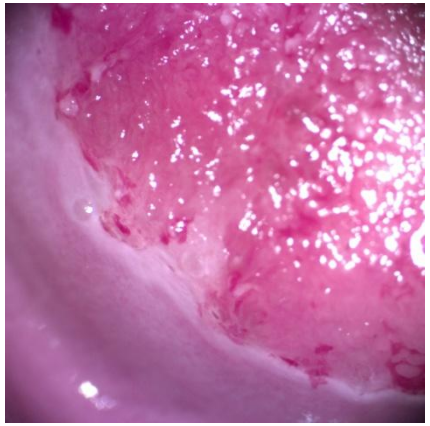
Normal Colposcopic appearances and features

Columnar Epithelium





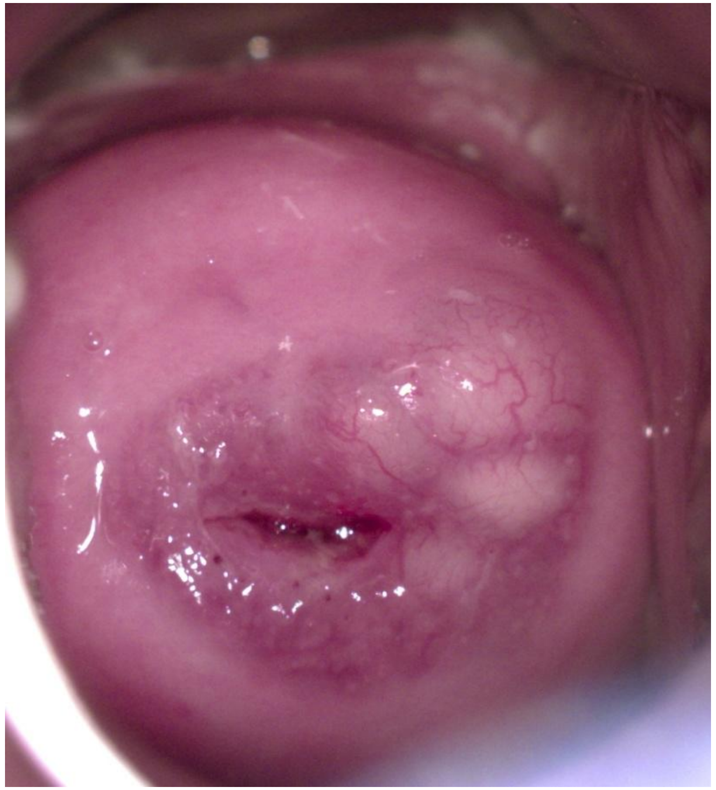


Mosaic and Punctation again...

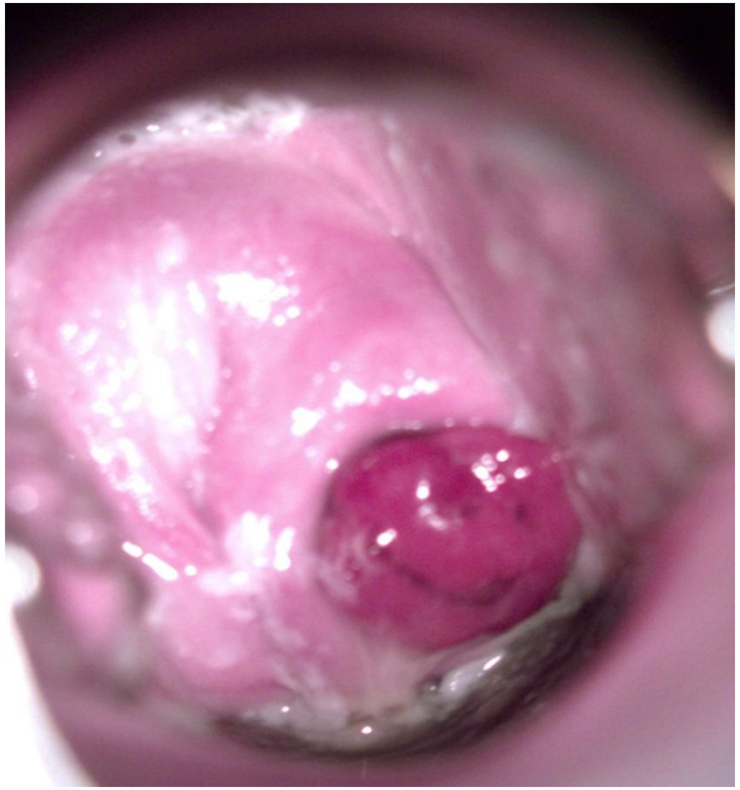
Mosaicism

Course Mosaicsim in CIN3

Nabothian follicle / Cyst



Endocervical Polyp



Invasive lesions

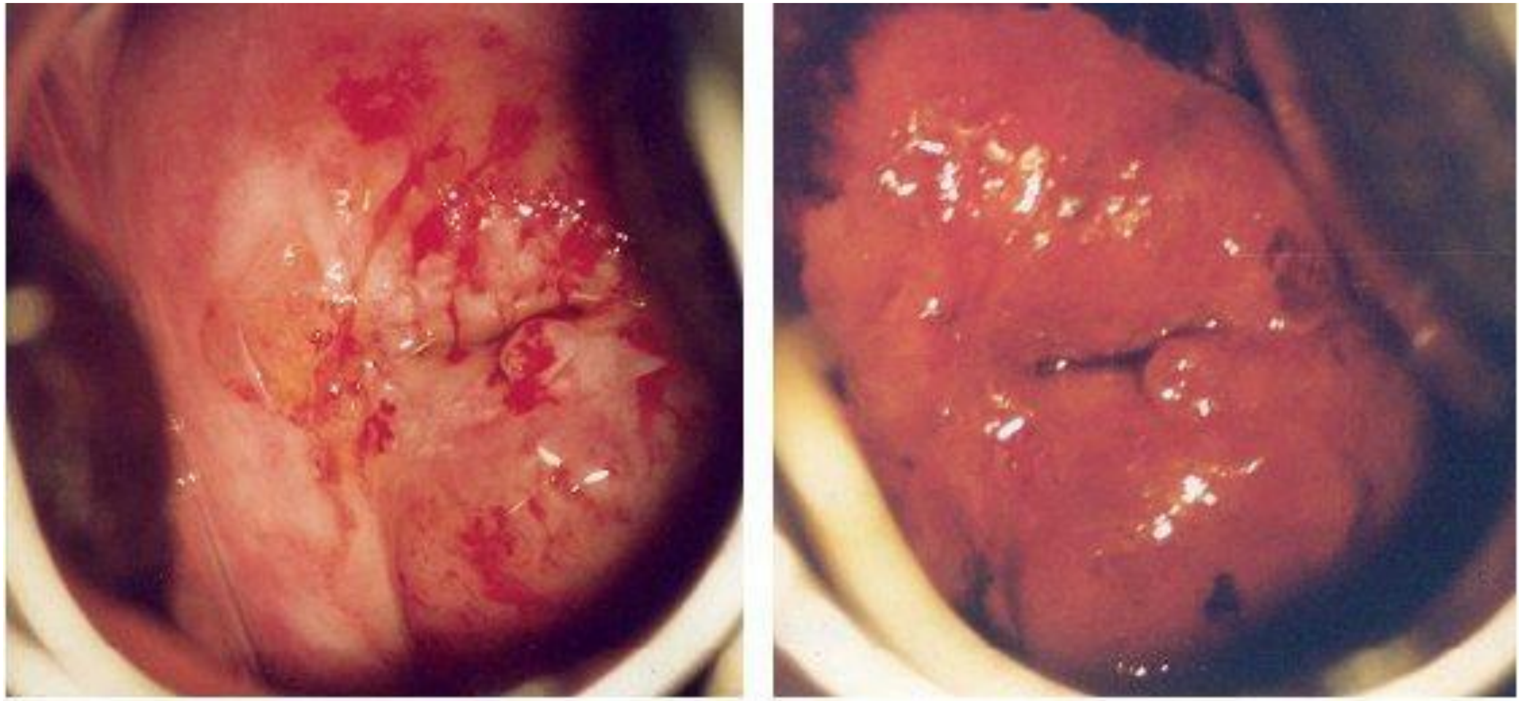


FIGURE 8.7: Invasive cervical cancer: (a) note the irregular surface contour with mountains-and-valleys appearance with atypical blood vessels in the dense acetowhite area; (b) appearance after the application of Lugol's iodine.

The biology of HPV

- Human papillomavirus (HPV) is a double-stranded DNA virus containing only eight genes. It infects epithelial cells and is usually transmitted by sexual contact.
- There are more than 100 subtypes of HPV:
- **low risk** – HPV 6 and 11, which can cause anogenital warts and respiratory papillomatosis
- **high risk** – HPV 16 and 18, which can cause cervical, anogenital and head and neck cancers. Other high risks are 31, 33, 35, 39, 45, 51, 52, 56, 58, 59 and 68. HPV 16, 18 are responsible for 70% of cervical cancers. Currently, the DNA amplication tests used in the NHS only differentiate between 'high-risk' and 'low-risk' HPVs varieties.
- HPV infection is endemic within the population.
- HPV proteins E6, E7 interact with cell proteins p53, pRb.
- In total, 99% of cervical cancers are estimated to be owing to HPV. Note that most women build up an immunity to the virus. If this was not the case, exposure to HPV would lead to cervical carcinoma in all women.

HPV vaccine

- A vaccine against HPV was developed following discovery of virus-like particles (VLPs). VLPs look exactly like virions but do not contain DNA and, therefore, have no infective properties. Type-specific antibodies, which neutralise infection, are produced in response to VLPs.
- In the UK, HPV vaccination (Cervarix®) was introduced into the National Immunisation Programme in September 2008. In September 2012, Gardasil®replaced Cervarix® in the UK HPV immunisation programme.
- Available vaccines:
- Gardasil® – quadrivalent (6, 11, 16 and 18)
- Cervarix® – bivalent (16 and 18); used in the UK.

Key points

- Human papilloma virus is a small DNA virus of 8000 pairs.
- It is usually transmitted via sexual intercourse. Risk factors for HPV infection include sexual behaviour, smoking, nutritional factors and immunosuppression.
- The two major oncogenes are E6 and E7. They produce proteins which interfere with tumour-suppressor genes monitoring the cell cycle.
- There are more than 100 different subtypes with different oncogenic potential (low-risk and high-risk types).
- High-risk types, HPV 16 and HPV 18, are associated with approximately 70% of cervical cancers.
- Persistent HPV infection from high-risk types are related to progression to high-grade dysplasia and cancer.

Thank you